Bayesian Optimization

Bayesian optimization is a methodology for optimizing expensive objective functions that has proven success in the sciences, engineering, and beyond. This timely text provides a self-contained and comprehensive introduction to the subject, starting from scratch and carefully developing all the key ideas along the way. This bottom-up approach illuminates unifying themes in the design of Bayesian optimization algorithms and builds a solid theoretical foundation for approaching novel situations.

The core of the book is divided into three main parts, covering theoretical and practical aspects of Gaussian process modeling, the Bayesian approach to sequential decision making, and the realization and computation of practical and effective optimization policies.

Following this foundational material, the book provides an overview of theoretical convergence results, a survey of notable extensions, a comprehensive history of Bayesian optimization, and an extensive annotated bibliography of applications.

Roman Garnett is Associate Professor in Computer Science and Engineering at Washington University in St. Louis. He has been a leader in the Bayesian optimization community since 2011, when he cofounded a long-running workshop on the subject at the NeurIPS conference. His research focus is developing Bayesian methods – including Bayesian optimization – for automating scientific discovery.

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